SIMATS ENGINEERING
DATA WAREHOUSING AND DATA MINING

COURSE: CLCSA16 DWDM **TOPIC:** Classification of Data Mining Systems

CLASSROOM ACTIVITY
SHORT ANSWER TEST

QUESTIONS: 10 **DURATION:** 30 minutes

STUDENT DETAILS
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Course: CSA 1614 - Data Warehousing and Data Mining

Assessment format: Short Answer Test

ANSWERS

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1. Define a Data Mining System.

A data mining system is an integrated set of tools and processes that analyzes large data collections to discover valid, useful, novel, and understandable patterns. It commonly includes data cleaning and integration, a mining engine, pattern evaluation, a knowledge base, and a user interface.

2. List the different classifications of Data Mining Systems.

Data mining systems are commonly classified by: (1) the type of data or database mined; (2) the type of knowledge discovered; (3) the techniques used; (4) the application domain; and (5) the degree or style of user interaction. These dimensions can be combined to describe one system.

3. Explain classification based on the type of data mined.

Data type	Description / example
Relational and transactional	Tables of entities and transactions, such as customer records or purchases.
Data warehouse	Integrated historical data organized across dimensions and measures.
Spatial, temporal, and sequence	Locations, time-stamped observations, event sequences, and time series.
Text and multimedia	Documents, web pages, images, audio, and video.
Web, graph, and heterogeneous data	Web logs and pages, linked entities, and data combined across different formats or systems.

The data type affects preprocessing, representation, and the mining algorithms that are appropriate. For example, spatial mining considers geographic relationships, while text mining first represents language as analyzable features.

4. Describe classification based on the type of knowledge mined.

A system may be designed to discover one or more of the following kinds of knowledge:

Knowledge type	Purpose
Characterization / discrimination	Summarizes a target class or compares it with another class.
Association / correlation	Finds items or attributes that occur together, such as products frequently purchased in one transaction.
Classification / prediction	Learns from examples to assign a class or estimate an unknown or future value.
Clustering	Groups similar records when class labels are not supplied.
Outlier analysis	Finds records that differ substantially from the general pattern.
Evolution / trend analysis	Studies changes, sequences, periodicity, and deviations over time.

5. Identify techniques used in Data Mining Systems.

Common techniques include database and data warehouse methods; statistics; machine learning; decision trees and rule induction; neural networks; pattern recognition; information retrieval and text processing; data visualization; and soft-computing methods such as fuzzy logic and genetic algorithms. A system may combine several techniques.

6. Explain classification based on applications.

Systems are grouped by the problem and domain they support. Examples include retail and marketing (customer segmentation and recommendations), banking and insurance (credit and fraud analysis), healthcare (risk and outcome analysis), telecommunications (churn and network faults), manufacturing (quality and predictive maintenance), cybersecurity (intrusion detection), and science (bioinformatics and climate analysis).

7. Describe classification based on user interaction.

An **interactive** or query-driven system lets users choose data, set constraints, adjust parameters, and explore results iteratively. An **autonomous** system runs predefined analyses with limited user input and presents discovered patterns or alerts. Many practical systems combine both approaches: automated discovery with user-directed follow-up.

8. Illustrate a real-world application in healthcare.

Illustrative application - hospital readmission risk: A hospital can combine historical encounter records with diagnoses, age group, prior admissions, length of stay, laboratory indicators, and follow-up information. Using past cases labeled as readmitted or not readmitted within a defined period, a classification model learns patterns and assigns a risk category to new discharges. Care teams can review high-risk cases for follow-up planning. The model should be validated for accuracy and bias, protect patient privacy, and support rather than replace clinical judgment.

9. Define classification in data mining.

Classification is a supervised learning task that uses a training set of records with known class labels to build a model, then uses that model to assign predefined categories to new records. Examples include classifying a transaction as fraudulent or legitimate, or a message as spam or not spam.

10. Differentiate between classification and clustering.

Aspect	Classification	Clustering
Learning type	Supervised	Unsupervised
Input labels	Known class labels are provided in training data.	No class labels are provided.
Goal	Predict the predefined class of a new record.	Discover groups of similar records in the data.
Typical output	A trained classifier and predicted class labels.	Clusters and their membership; group meanings are interpreted afterward.
Example	Classify a claim as valid or potentially fraudulent.	Group patients with similar observed characteristics.

In short: classification predicts known categories; clustering discovers structure without predefined categories.